directory via Identity Center), the comparison stays consistent and doesn't need policy updates when individuals change departments.

8.4.3 Applying the PARC model to AWS IAM

IAM was built specifically for the AWS ecosystem, so its interpretation of Principal, Action, Resource, and Context is closely tied to AWS services and infrastructure. Unlike general-purpose policy languages, IAM can make strong assumptions about these elements because they directly correspond to AWS concepts.

- **Principal**—Principals in IAM are AWS identities, including IAM users, IAM roles, or federated identities from sources like Identity Center. Attributes can be added to principals as tags, which can be referenced in policies using `aws:PrincipalTag/<attribute-name>`. This setup allows external identity attributes, such as those from ACME's corporate directory, to be incorporated directly into access decisions.
- **Action**—Actions in IAM are closely linked to AWS API operations, shown as strings like `"s3:GetObject"` or `"ec2:StartInstances`." Each AWS service has its own set of actions, enabling IAM policies to be very specific about which operations are allowed or denied.
- **Resource**—Resources are AWS objects identified by Amazon Resource Names (ARNs), such as `arn:aws:s3:::acme-docs/report.pdf`. Many services support resource-based policies, which allow permissions to be granted directly on a resource in addition to identity-based policies attached to principals. Resource tagging is deeply integrated into the model and is often used for attribute-based conditions.
- **Context**—IAM supports extensive context evaluation using condition keys. These include AWS global keys (like `aws:CurrentTime` or `aws:SourceIp`), service-specific keys (which have a prefix like `s3:`), and custom principal or resource tags. Conditions are represented in JSON within the `Condition` element of a policy statement, making context checks a key part of policy evaluation.